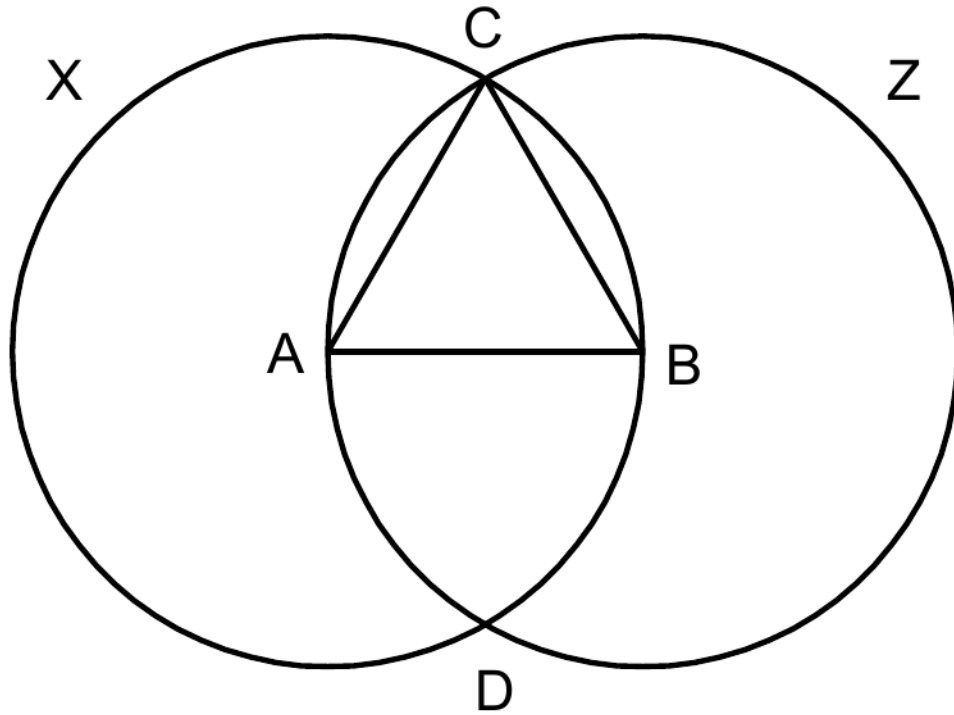


THEOREM 1

How to make an equilateral triangle

Given a straight line AB, construct an equilateral triangle on it so that AB is the base.

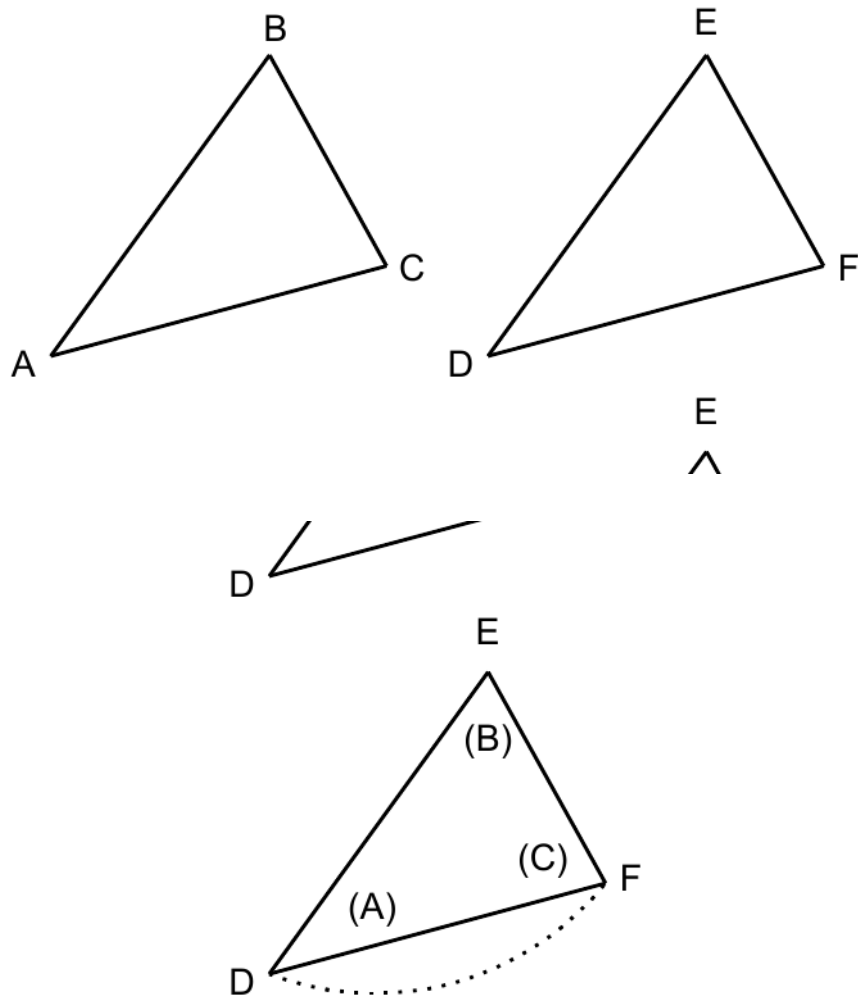


THEOREM 2

THEOREM

Side-Angle-Side (SAS)

If in one triangle a side, the next angle, and the next side are respectively equal to a side, the next angle, and the next side in another triangle, then all the corresponding sides and angles of the two triangles are equal, and they have equal areas.

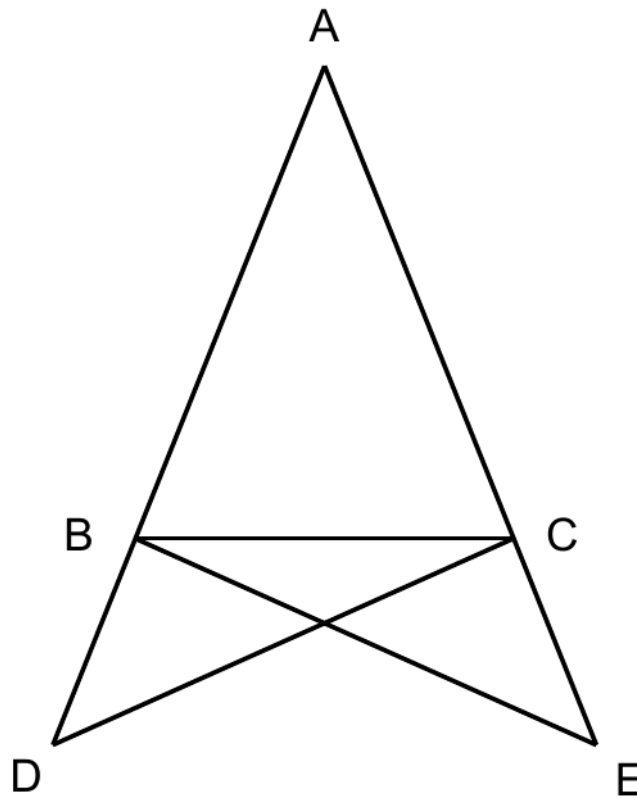


THEOREM 3

THEOREM

Base angles of an isosceles triangle

In an isosceles triangle, the base angles are equal to each other, and the angles under the base are equal to each other.

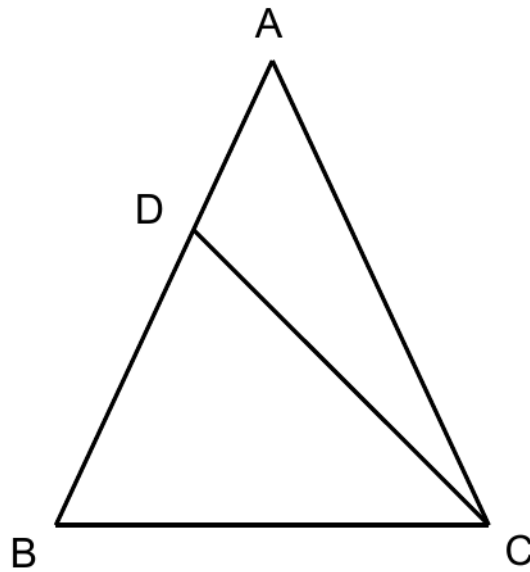


THEOREM 4

THEOREM

Converse of the isosceles base-angle theorem

If two angles in a triangle are equal to each other, then the sides opposite them are also equal — the triangle is isosceles (at least two sides equal).

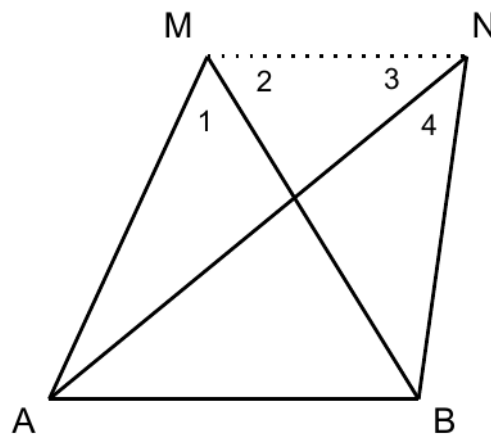
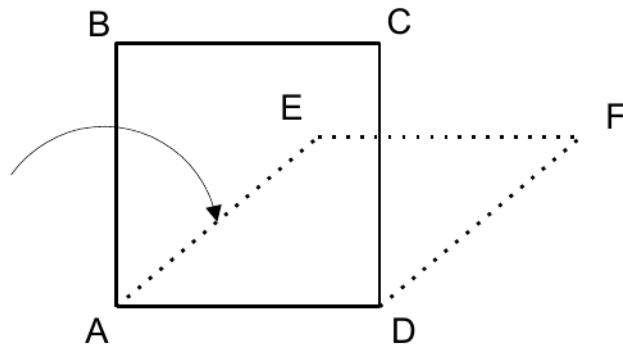


THEOREM 5

THEOREM

Triangles are rigid

Triangles are rigid: given three straight lines making a triangle, you cannot change the angles while keeping all three sides the same lengths.

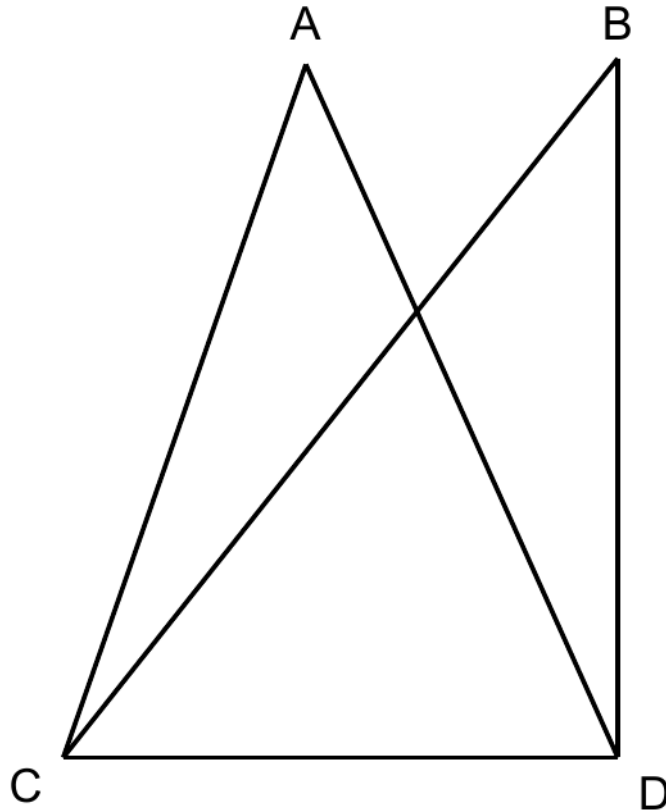


THEOREM 6

THEOREM

Side-Side-Side (SSS)

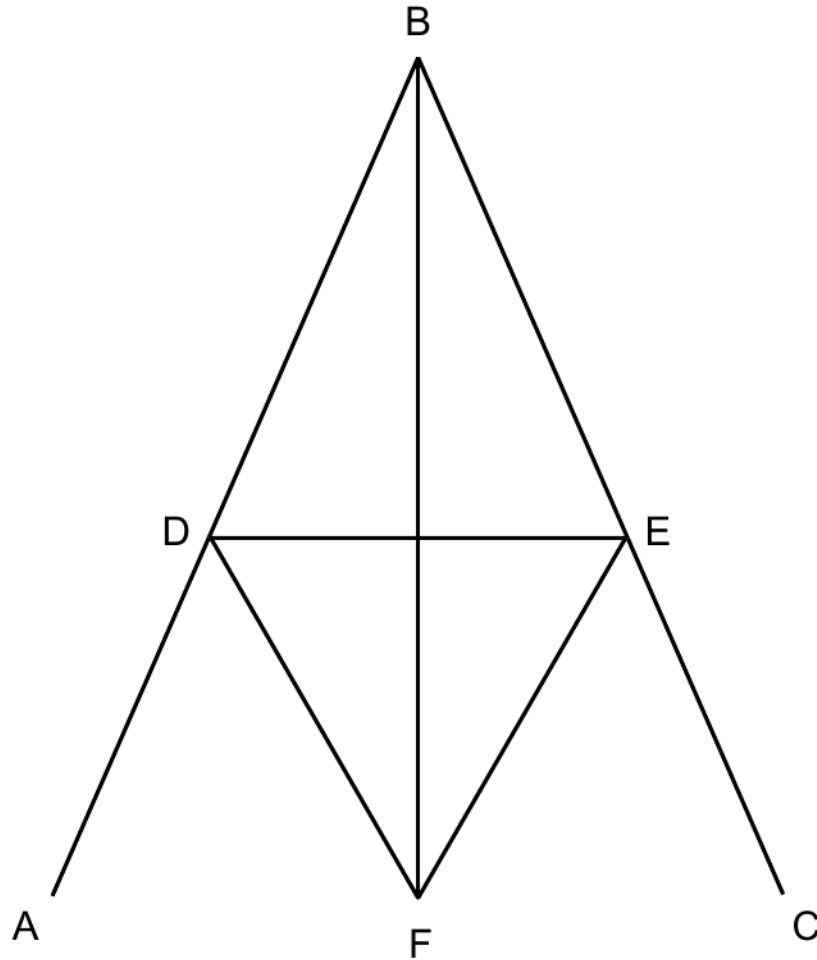
If the three sides of one triangle are equal to the three corresponding sides of another triangle, then the two triangles also have their corresponding angles equal (those between equal sides), and equal areas.



THEOREM 7

How to bisect any rectilineal angle

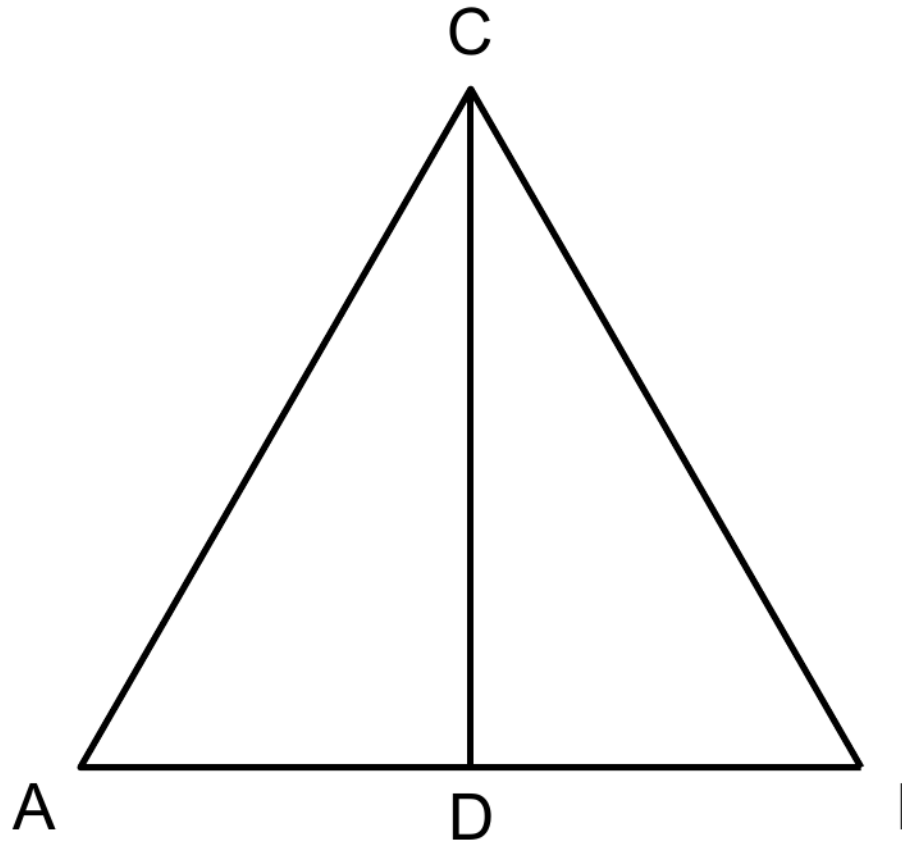
Given any rectilineal angle ABC, construct the ray that bisects it.



THEOREM 8

How to bisect any straight line

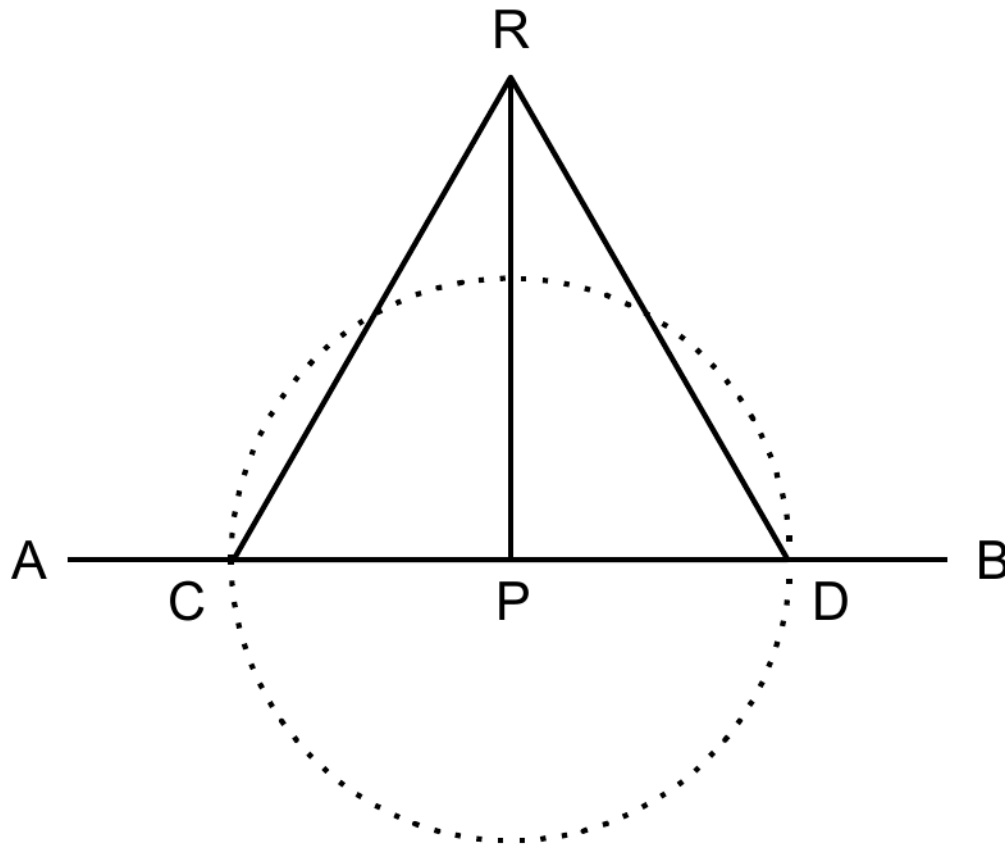
Given a straight line AB, construct its midpoint.



THEOREM 9

Erect a perpendicular from a point on a line

Given a straight line AB and a point P on it, draw a line from P at right angles to AB.

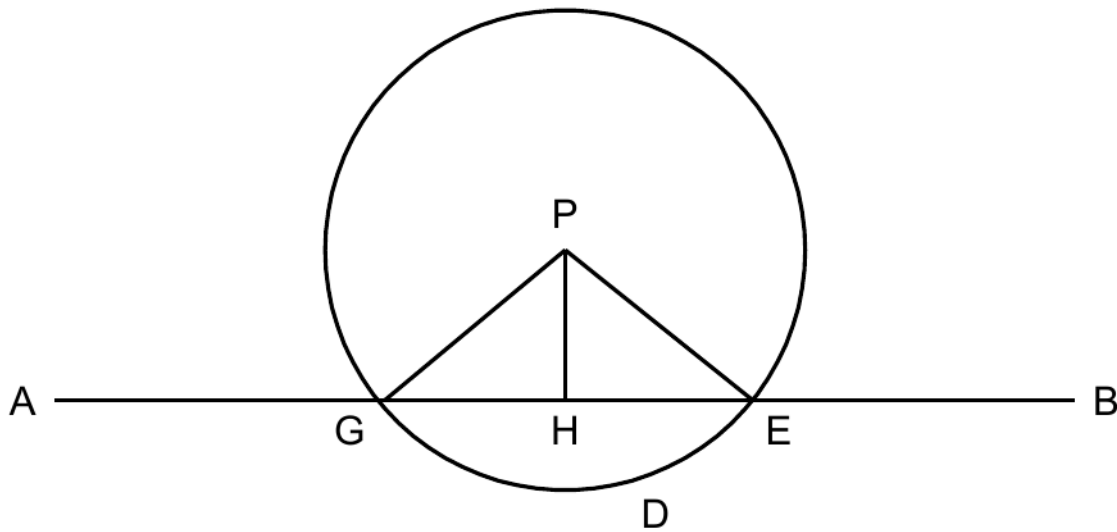


THEOREM 10

CONSTRUCTION

Drop a perpendicular from a point to a line

Given a straight line AB and a point P not on it, drop a perpendicular from P to AB.

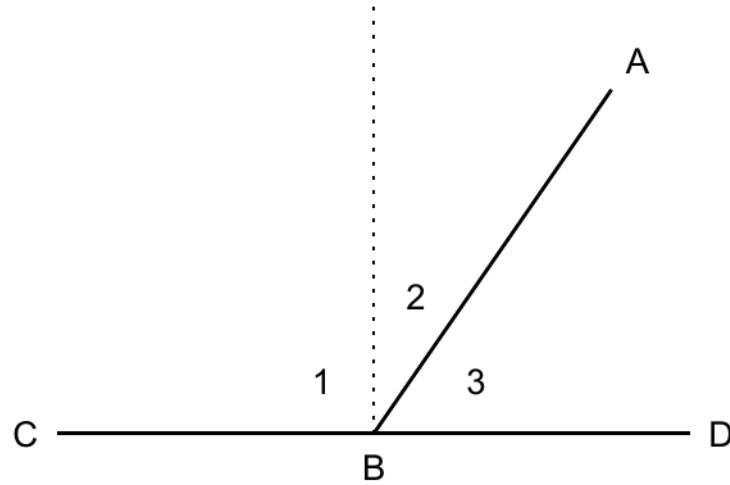


THEOREM 11

THEOREM

Adjacent angles on a straight line

When one straight line stands on another, the adjacent angles add up to two right angles.

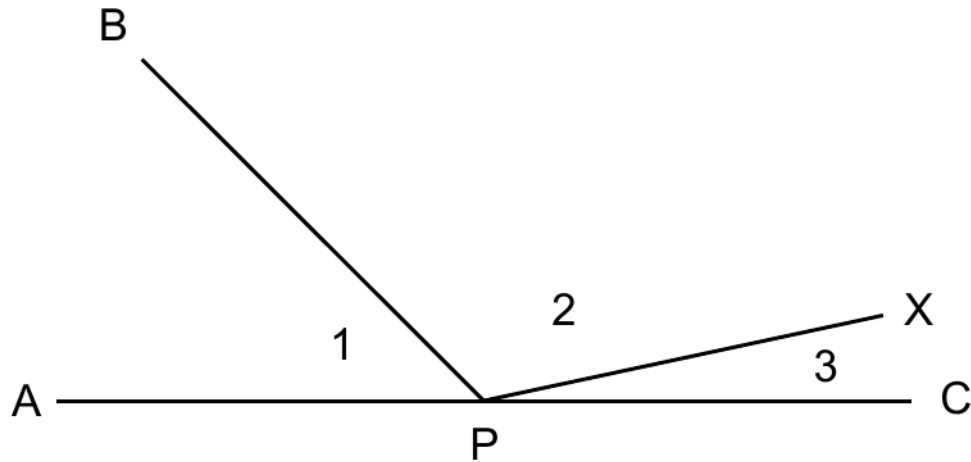


THEOREM 12

THEOREM

Converse: angles sum to two rights $\Rightarrow$ straight line

If two adjacent angles add up to two right angles, then the outer legs form one straight line.

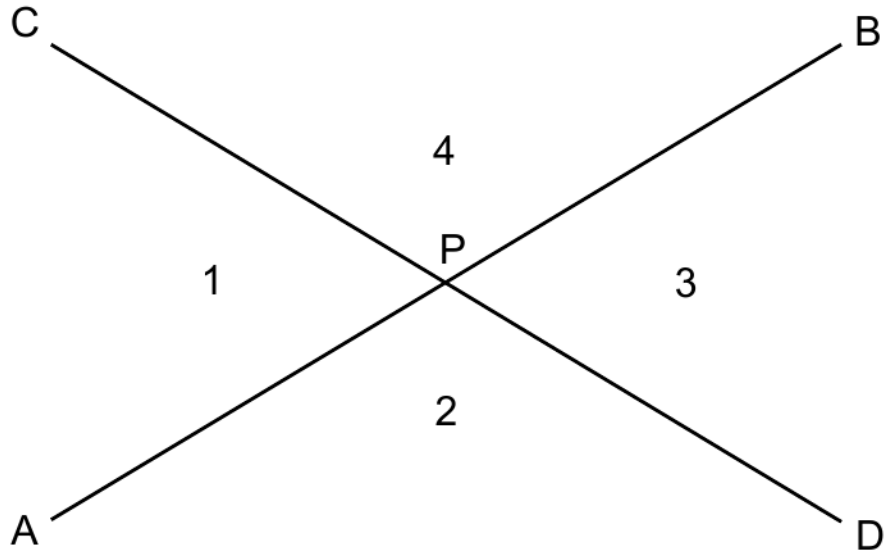


THEOREM 13

THEOREM

Vertical angles are equal

When two straight lines cut each other, they make the vertical (opposite) angles equal.

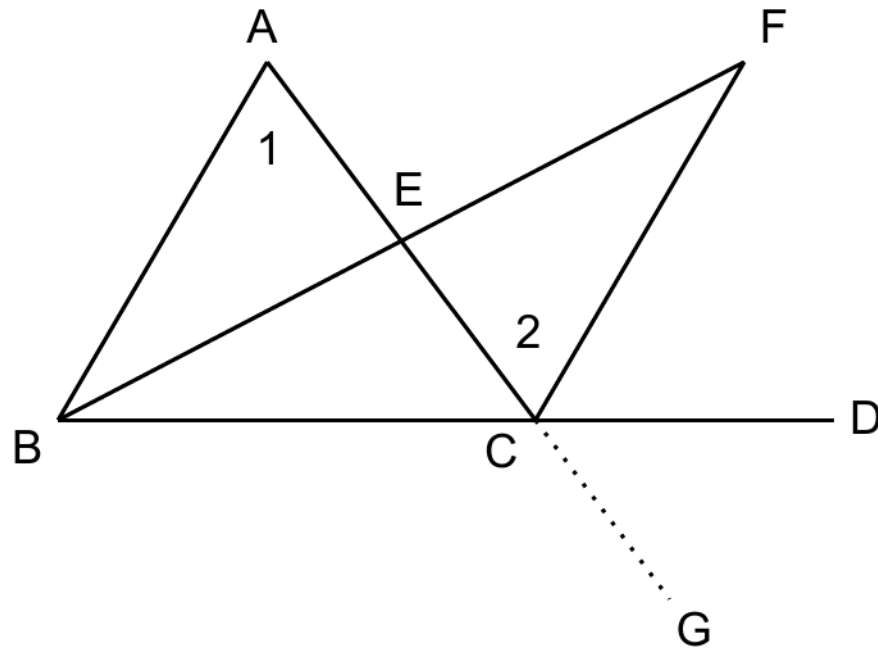


THEOREM 14

THEOREM

Exterior angle greater than remote interior

If any side of a triangle is extended, the exterior angle is greater than either of the interior and opposite (remote) angles.



[1] Bisect AC at E.

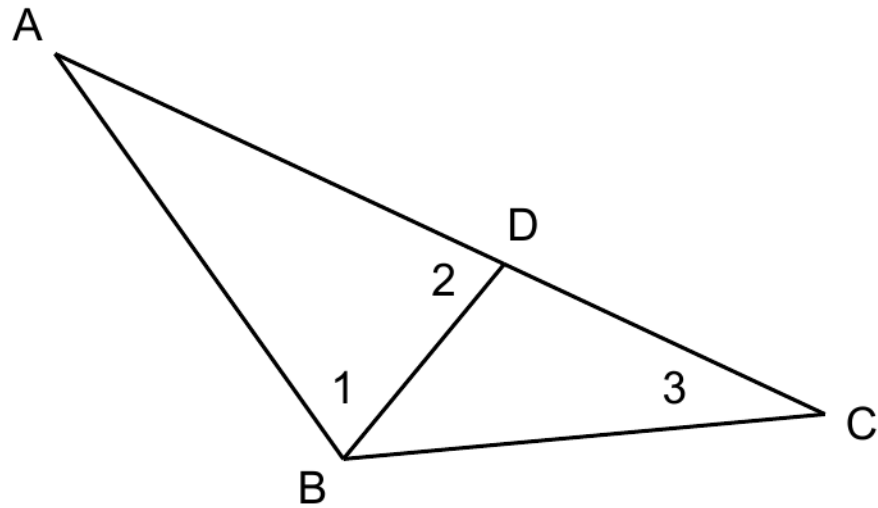
[2] Join BE and produce it for enough

THEOREM 15

THEOREM

Greater side opposite greater angle

In any triangle, a greater side has opposite to it a greater angle.

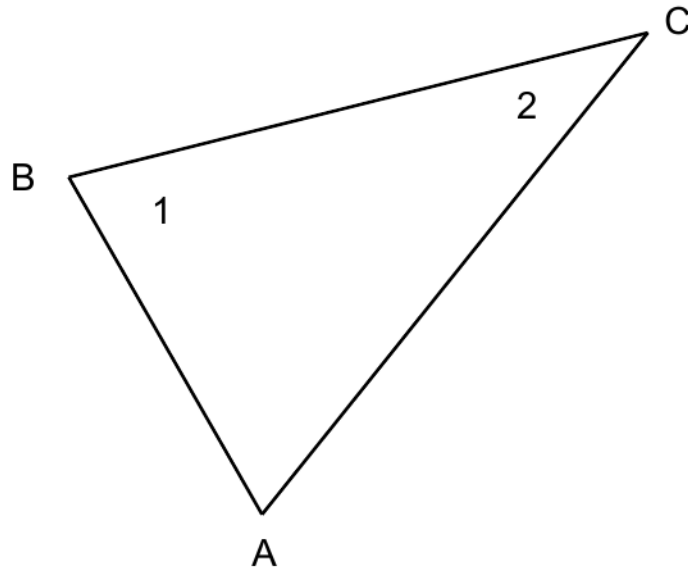


THEOREM 16

THEOREM

Greater angle opposite greater side

In any triangle, a greater angle has opposite to it a greater side.

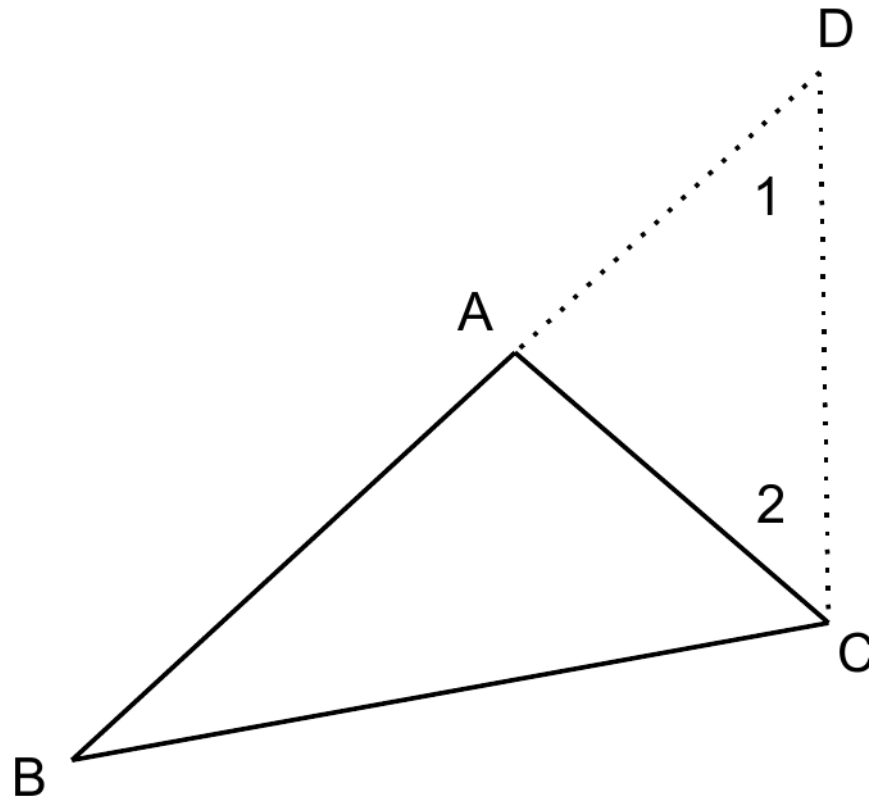


THEOREM 17

THEOREM

Triangle inequality

In any triangle, any side is less than the sum of the other two sides, but greater than their difference.

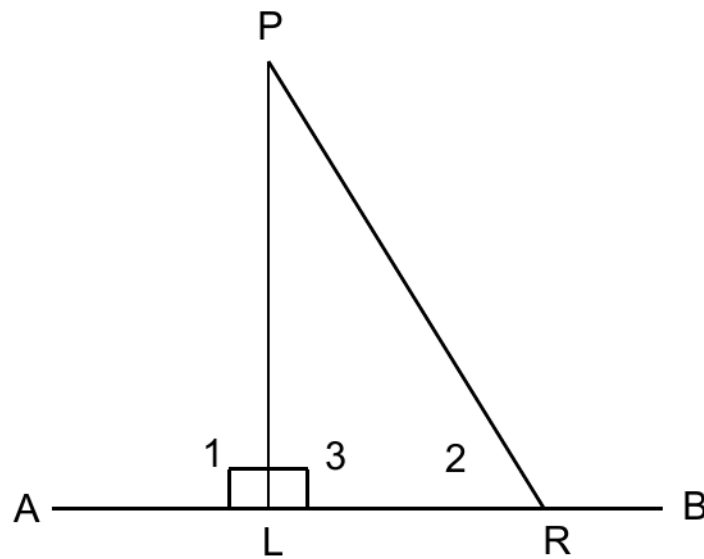


THEOREM 18

THEOREM

Shortest path to a line is the perpendicular

The shortest distance from a point to a straight line is the perpendicular from the point to the line.

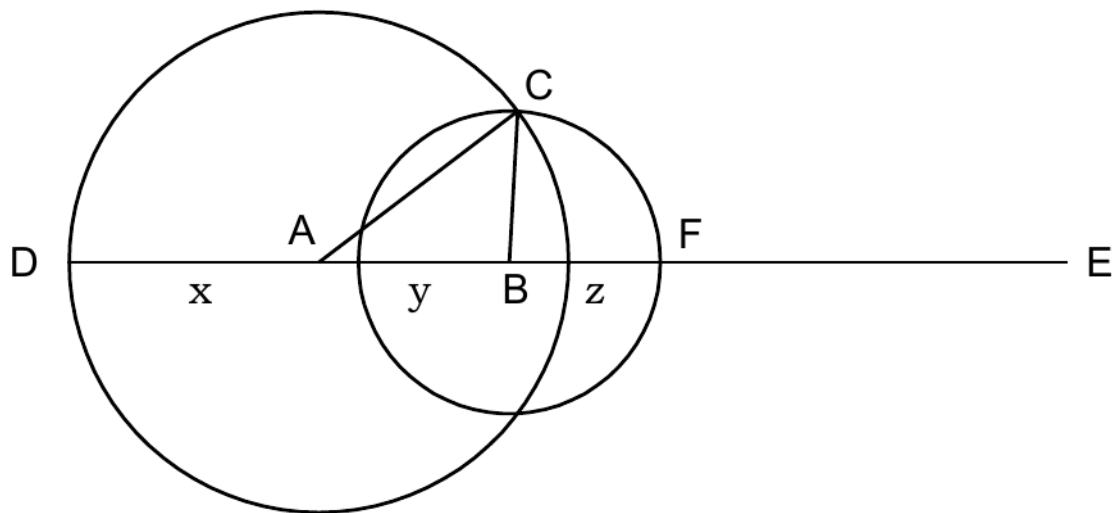


straight lines is shortest? The p
line drawn from P to any random

THEOREM 19

Construct a triangle from three sides

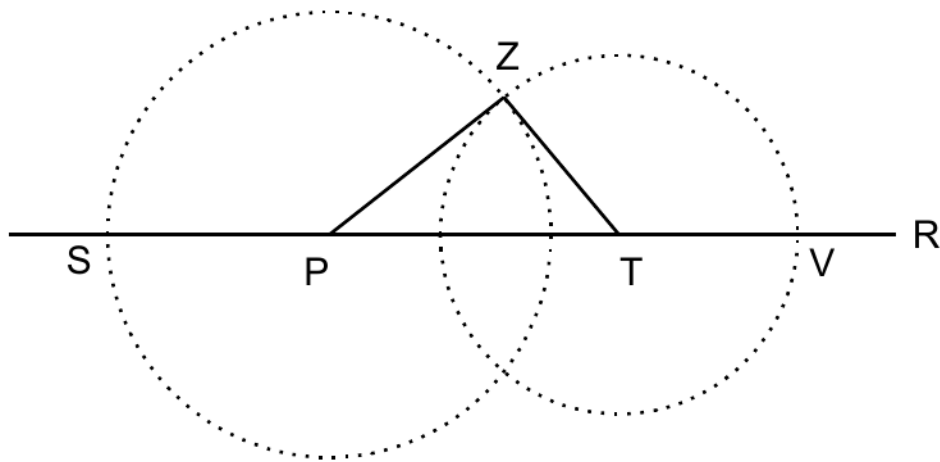
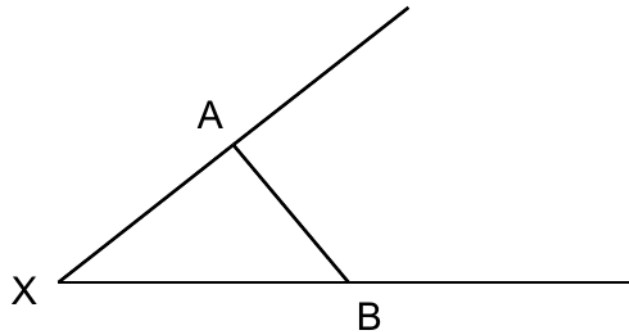
Given three straight lines X , Y , Z such that any two together are greater than the third, construct a triangle with those side lengths.



THEOREM 20

Copy an angle

Given an angle X and a ray PR , construct an angle at P , with one side along PR , equal to angle X .



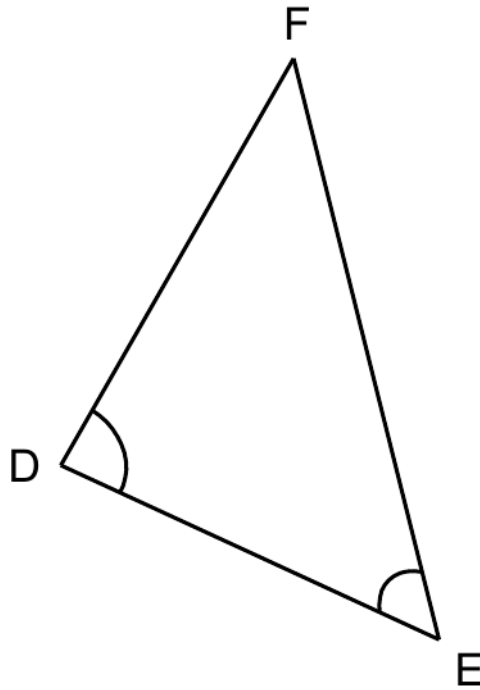
[2] Extend RP , and
cut off $PS = XA$
cut off $DT = YR$

THEOREM 21

THEOREM

Angle-Side-Angle (ASA)

If in a triangle two angles and the side joining them are respectively equal to two angles and the side joining them in another triangle, then all corresponding sides and angles are equal, and the triangles have equal area.

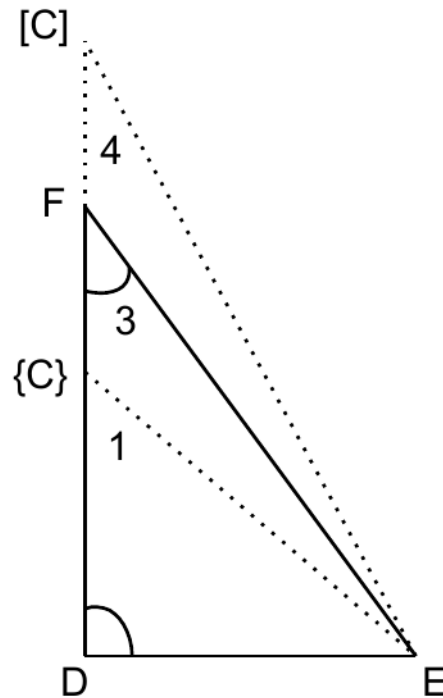


THEOREM 22

THEOREM

Angle-Angle-Side (AAS)

If two triangles have two angles equal to two angles, and a side equal to a corresponding side opposite an equal angle, then they have all corresponding sides and angles equal, and equal areas.

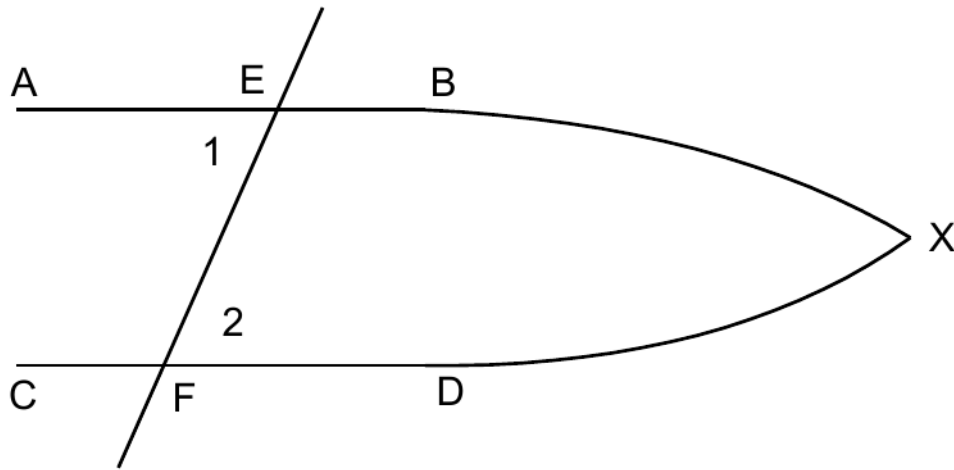


THEOREM 23

THEOREM

Equal alternate angles $\Rightarrow$ parallel

If two straight lines are cut by a third making the alternate angles equal, then the two straight lines are parallel.

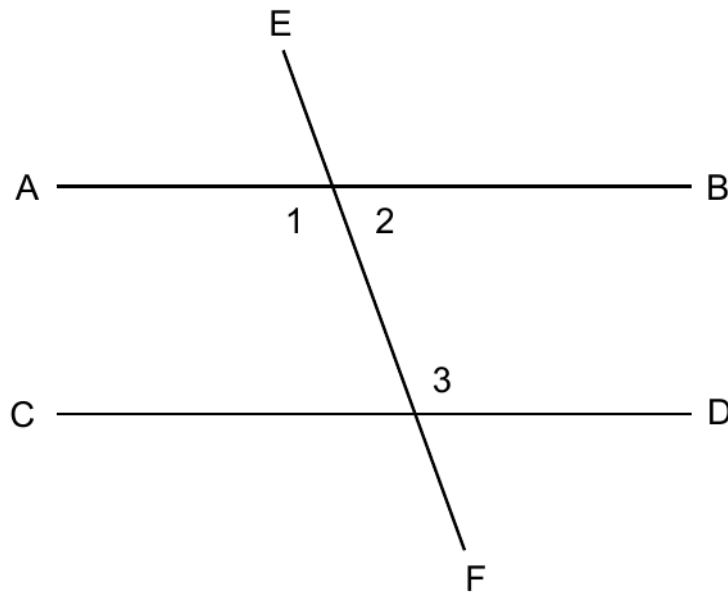


THEOREM 24

THEOREM

Co-interior angles supplementary $\Rightarrow$ parallel

If two straight lines are cut by a third making interior angles on the same side add up to two right angles, then the two lines are parallel.

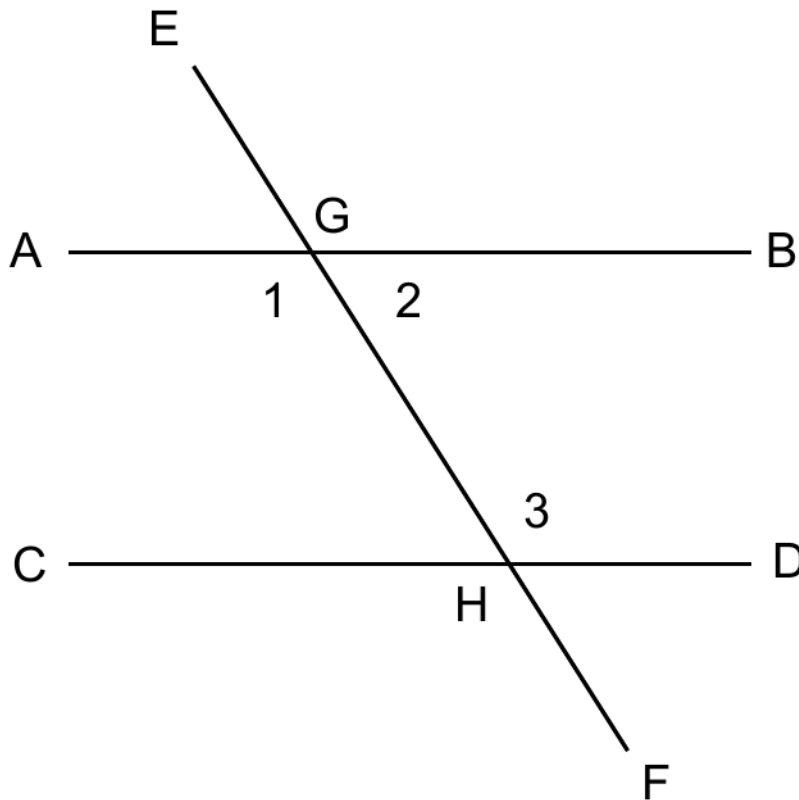


THEOREM 25

THEOREM

Parallel $\Rightarrow$ alternate equal and co-interior supplementary

If a straight line cuts two parallels, it makes alternate angles equal and interior angles on the same side supplementary (sum to two right angles).

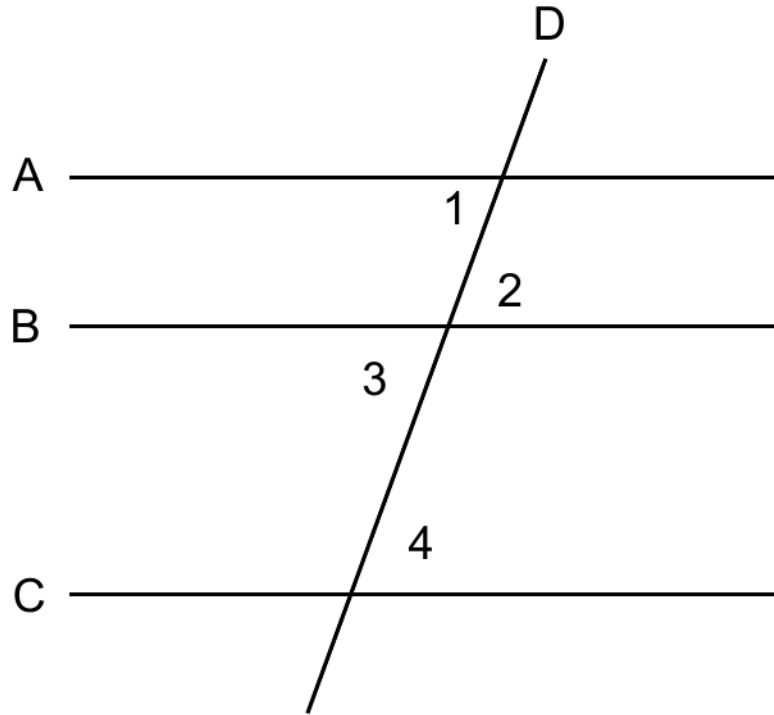


THEOREM 26

THEOREM

Transitivity of parallelism

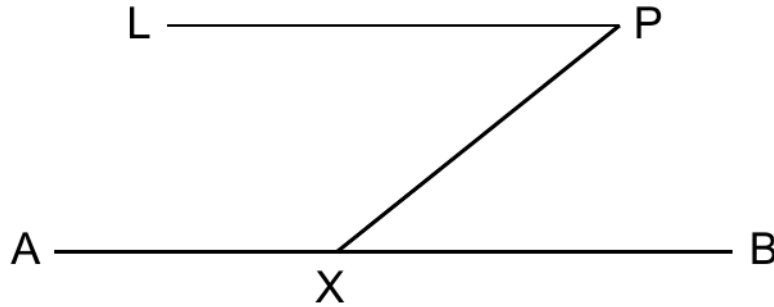
Straight lines parallel to the same straight line are parallel to each other.



THEOREM 27

Draw a parallel through a point

Given a line AB and a point P not on it, draw the line through P parallel to AB.

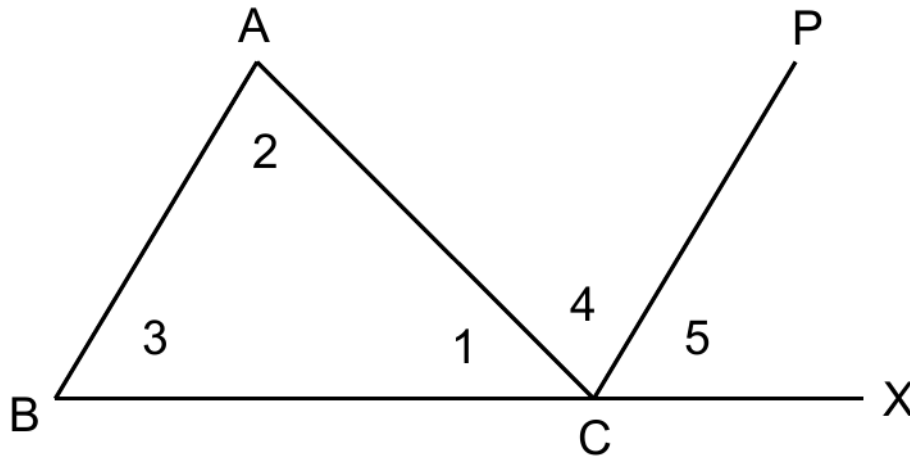


THEOREM 28

THEOREM

Angle sum of a triangle

In any triangle the three angles add up to two right angles, and an exterior angle equals the sum of the two remote interior angles.

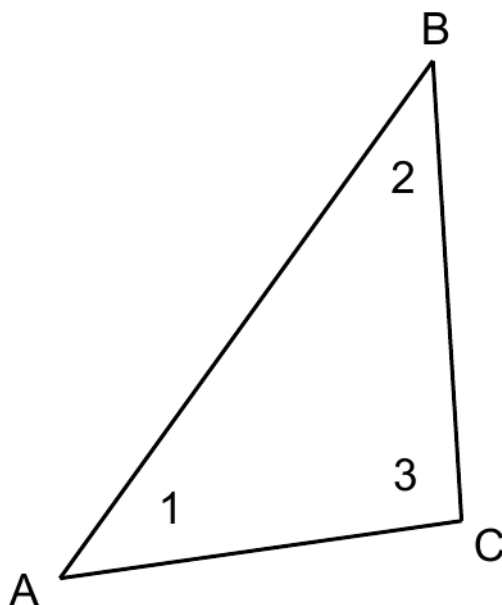


THEOREM 29

THEOREM

Third angle equal

If two triangles have two angles equal to two angles, then the remaining angles are equal.

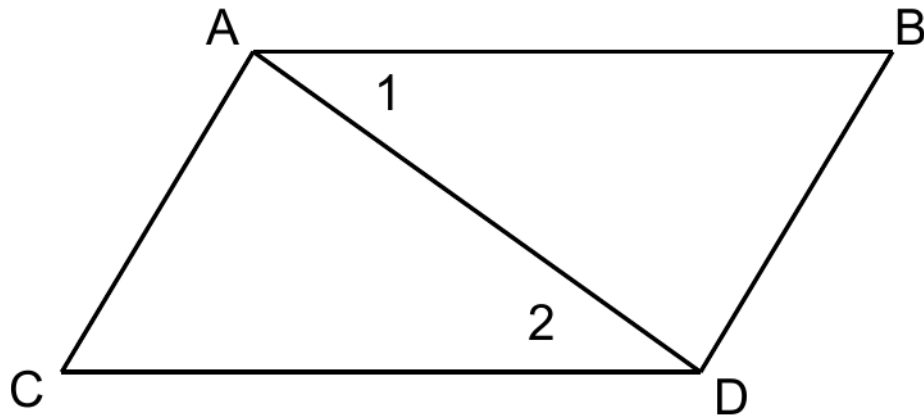


THEOREM 30

THEOREM

One pair parallel and equal $\Rightarrow$ parallelogram

If one pair of opposite sides in a quadrilateral are both parallel and equal, then the quadrilateral is a parallelogram (the other pair is also parallel and equal).

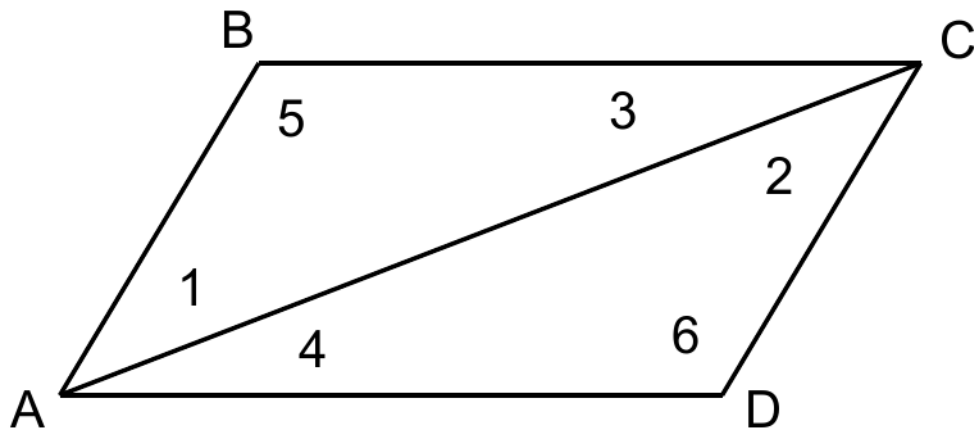


THEOREM 31

THEOREM

Properties of a parallelogram

In any parallelogram, opposite sides and opposite angles are equal, and a diagonal bisects the area of the parallelogram.

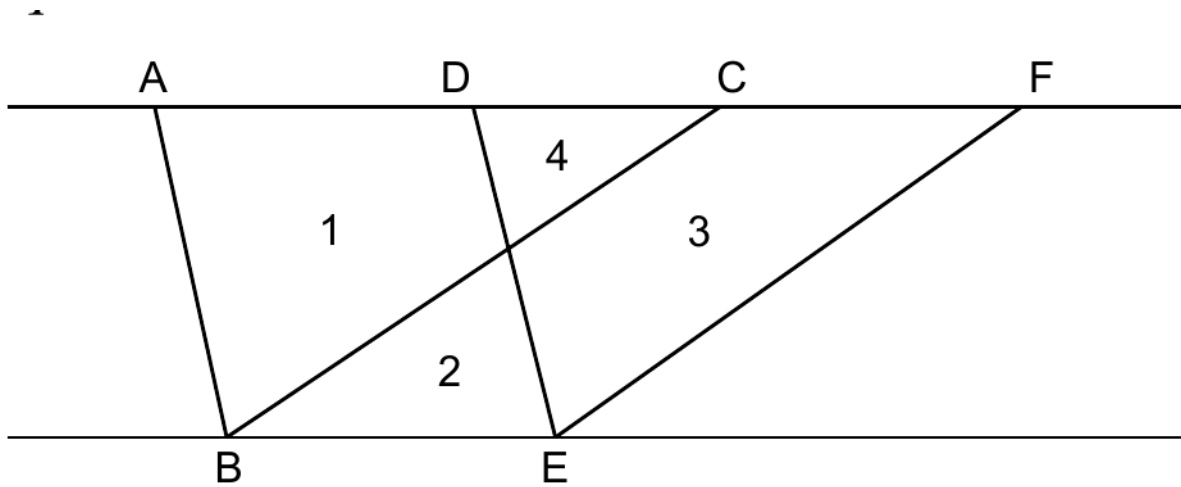


THEOREM 32

THEOREM

Parallelograms on the same base and in the same parallels

Parallelograms on the same base and in the same parallels have equal areas.

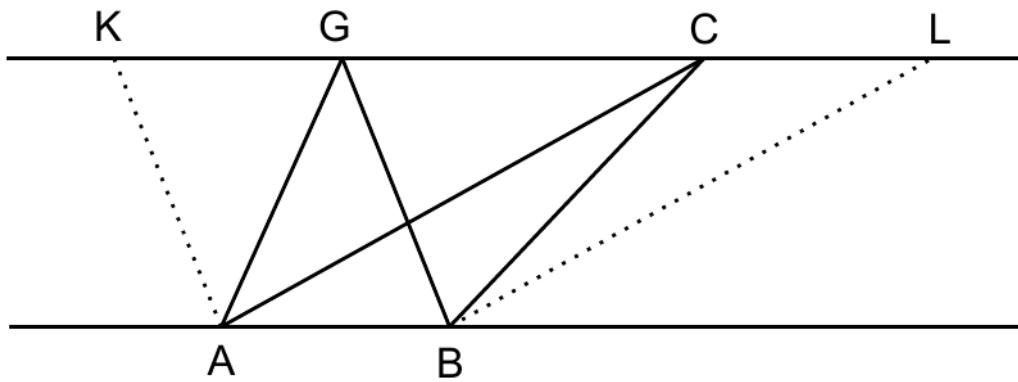


THEOREM 33

THEOREM

Triangles on the same base and in the same parallels

Triangles on the same base and in the same parallels have equal areas; a triangle on the same base and in the same parallels as a parallelogram has half the parallelogram's area.

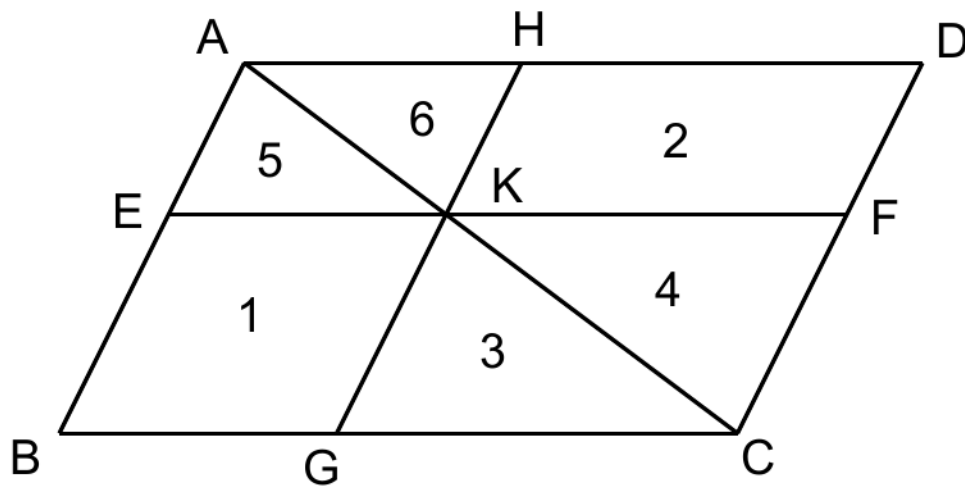


THEOREM 34

THEOREM

Complements of parallelograms about a diagonal

In any parallelogram, the “complements” of the parallelograms about a diagonal are equal in area.

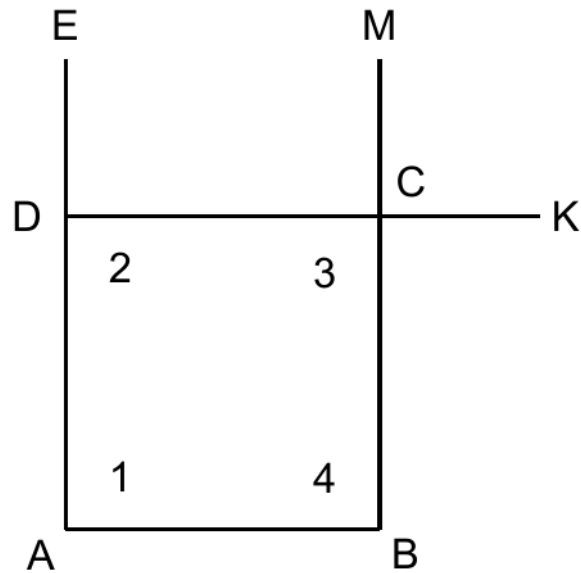


their own diagonals, and also forming two ‘
(or 1 & 2). These complementary parallelograms

THEOREM 35

How to make a square

Given a straight line AB, construct a square having AB as one side.

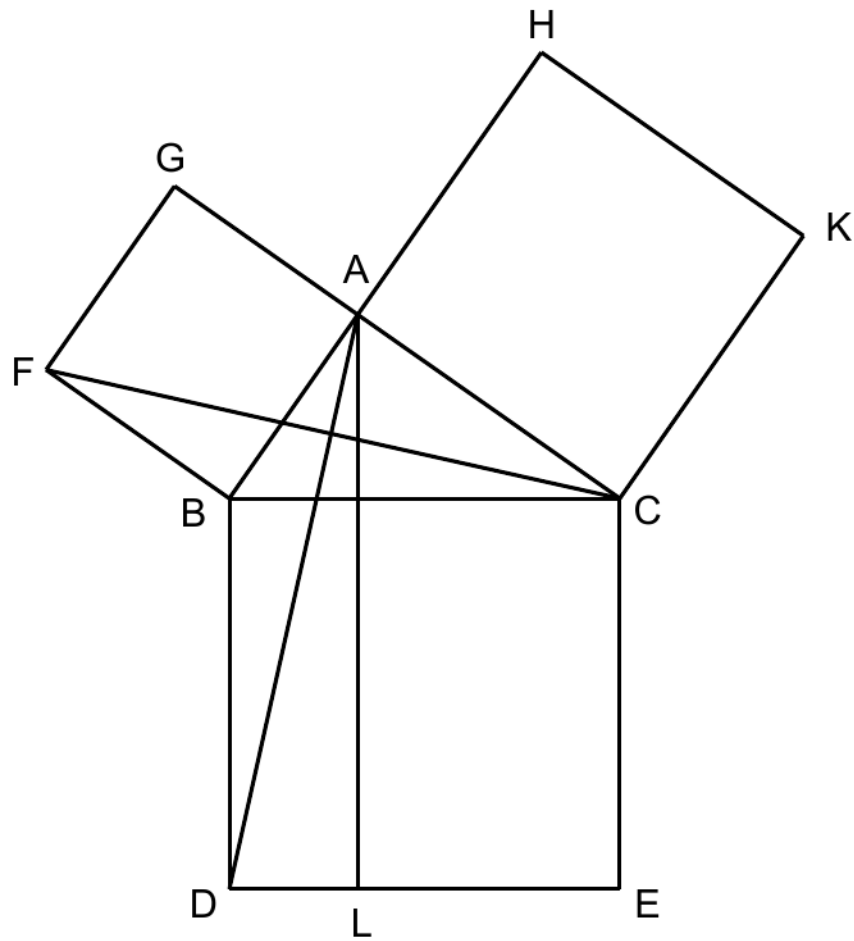


THEOREM 36

THEOREM

The Pythagorean Theorem

In any right triangle, the square on the hypotenuse is equal to the sum of the two squares on the remaining sides.



THEOREM 37

THEOREM

Converse of Pythagoras

If the square on one side of a triangle equals the sum of the squares on the other two sides, then the angle opposite that side is a right angle.

